

8/2/2026

Unfortunately, there is still a little more infrastructure that needs to be built up before I can get to constructing interesting R -algebras satisfying certain universal properties.

Theorem 2.1: Let $M_1, \dots, M_p$ be R -modules (where $p > 2$) and suppose $1 \leq q < p$. Then there is an R -module isomorphism from $(\bigotimes_{i=1}^q M_i) \otimes (\bigotimes_{i=q+1}^p M_i)$ to $\bigotimes_{i=1}^p M_i$ such that $(m_1 \otimes \dots \otimes m_q) \otimes (m_{q+1} \otimes \dots \otimes m_p) \mapsto m_1 \otimes \dots \otimes m_p$ for all $m_i \in M_i$.

Proof:

I shall proceed by induction on p . For our base case of $p = 3$, recall from [pages 1067-1069](#) that my original construction of $M_1 \otimes M_2 \otimes M_3$ was the set $M_1 \otimes (M_2 \otimes M_3)$ where $m_1 \otimes m_2 \otimes m_3$ is equal to $m_1 \otimes (m_2 \otimes m_3)$. Meanwhile, I showed on [pages 730-731](#) that $M_1 \otimes (M_2 \otimes M_3)$ is isomorphic to $(M_1 \otimes M_2) \otimes M_3$ via an isomorphism mapping $m_1 \otimes (m_2 \otimes m_3)$ to $(m_1 \otimes m_2) \otimes m_3$. Hence, our base case is entirely dealt with via prior results.

Next suppose $p > 3$.

- If $q = 1$ then recall that $\bigotimes_{i=2}^p M_i$ is isomorphic to $M_2 \otimes (\dots \otimes (M_{p-1} \otimes M_p) \dots)$ via an isomorphism ϕ such that:

$$m_2 \otimes \dots \otimes m_{p-1} \otimes m_p \mapsto m_2 \otimes (\dots \otimes (m_{p-1} \otimes m_p) \dots).$$

In turn, $\text{Id}_{M_1} \otimes \phi$ is an isomorphism from $M_1 \otimes (\bigotimes_{i=2}^p M_i)$ to $M_1 \otimes (M_2 \otimes (\dots \otimes (M_{p-1} \otimes M_p) \dots))$ such that:

$$m_1 \otimes (m_2 \otimes \dots \otimes m_p) \mapsto m_1 \otimes (m_2 \otimes (\dots \otimes (m_{p-1} \otimes m_p) \dots)).$$

And via the reasoning on [pages 1067-1069](#), we can compose that map with an isomorphism $\psi : M_1 \otimes (M_2 \otimes (\dots \otimes (M_{p-1} \otimes M_p) \dots)) \rightarrow \bigotimes_{i=1}^p M_i$ sending $m_1 \otimes (m_2 \otimes (\dots \otimes (m_{p-1} \otimes m_p) \dots))$ to $m_1 \otimes \dots \otimes m_p$.

- Meanwhile, if $q > 1$ then we can conclude by our induction hypothesis that $\bigotimes_{i=1}^q M_i$ is isomorphic to $M_1 \otimes \bigotimes_{i=2}^q M_i$ via an isomorphism ϕ mapping $m_1 \otimes \dots \otimes m_q$ to $m_1 \otimes (m_2 \otimes \dots \otimes m_q)$.

In turn, $\phi \otimes \text{Id}_{\bigotimes_{i=q+1}^p M_i}$ is an isomorphism from $(\bigotimes_{i=1}^q M_i) \otimes (\bigotimes_{i=q+1}^p M_i)$ to $(M_1 \otimes (\bigotimes_{i=2}^q M_i)) \otimes (\bigotimes_{i=q+1}^p M_i)$ such that $(m_1 \otimes \dots \otimes m_q) \otimes (m_{q+1} \otimes \dots \otimes m_p)$ is mapped to $(m_1 \otimes (m_2 \otimes \dots \otimes m_q)) \otimes (m_{q+1} \otimes \dots \otimes m_p)$.

Next, by the exercise on [pages 730-731](#) we have an isomorphism from $(M_1 \otimes (\bigotimes_{i=2}^q M_i)) \otimes (\bigotimes_{i=q+1}^p M_i)$ to $M_1 \otimes ((\bigotimes_{i=2}^q M_i) \otimes (\bigotimes_{i=q+1}^p M_i))$ sending $(m_1 \otimes (m_2 \otimes \dots \otimes m_q)) \otimes (m_{q+1} \otimes \dots \otimes m_p)$ to $m_1 \otimes ((m_2 \otimes \dots \otimes m_q) \otimes (m_{q+1} \otimes \dots \otimes m_p))$.

Then by our induction hypothesis, $(\bigotimes_{i=2}^q M_i) \otimes (\bigotimes_{i=q+1}^p M_i)$ is isomorphic to $\bigotimes_{i=2}^p M_i$ via an isomorphism ψ mapping $(m_2 \otimes \dots \otimes m_q) \otimes (m_{q+1} \otimes \dots \otimes m_p)$ to $m_2 \otimes \dots \otimes m_p$.

In turn, $\text{Id}_{M_1} \otimes \psi$ is an isomorphism from $M_1 \otimes ((\bigotimes_{i=2}^q M_i) \otimes (\bigotimes_{i=q+1}^p M_i))$ to $M_1 \otimes (\bigotimes_{i=2}^p M_i)$ sending $m_1 \otimes ((m_2 \otimes \dots \otimes m_q) \otimes (m_{q+1} \otimes \dots \otimes m_p))$ to $m_1 \otimes (m_2 \otimes \dots \otimes m_p)$.

Finally, we can now use our reasoning for the case when $q = 1$ to finish up. ■

Next let $A_1, \dots, A_p$ be R -algebras. I already mentioned how $\bigotimes_{i=1}^p A_i$ can naturally be turned into an R -algebra itself. However, here is another interpretation of this natural product operation on $\bigotimes_{i=1}^p A_i$.

Consider the $2p$ -linear map $T : A_1 \times \dots \times A_p \times A_1 \times \dots \times A_p \rightarrow \bigotimes_{i=1}^p A_i$ given by:

$$T(a_1, \dots, a_p, a'_1, \dots, a'_p) := (a_1 a'_1, \dots, a_p a'_p).$$

By the universal property of tensor products we thus know there is a unique R -module homomorphism $\phi : A_1 \otimes \dots \otimes A_p \otimes A_1 \otimes \dots \otimes A_p \rightarrow \bigotimes_{i=1}^p A_i$ such that:

$$\phi(a_1 \otimes \dots \otimes a_p \otimes a'_1 \otimes \dots \otimes a'_p) := (a_1 a'_1, \dots, a_p a'_p).$$

Then by the prior theorem, we can also say that there is a unique R -module homomorphism $\tilde{\phi} : (\bigotimes_{i=1}^p A_i) \otimes (\bigotimes_{i=1}^p A_i) \rightarrow \bigotimes_{i=1}^p A_i$ such that:

$$\tilde{\phi}((a_1 \otimes \dots \otimes a_p) \otimes (a'_1 \otimes \dots \otimes a'_p)) := (a_1 a'_1, \dots, a_p a'_p).$$

And finally, we can conclude that there is a unique bilinear map

$$\mu : (\bigotimes_{i=1}^p A_i) \times (\bigotimes_{i=1}^p A_i) \rightarrow \bigotimes_{i=1}^p A_i$$
 such that:

$$\mu((a_1 \otimes \dots \otimes a_p), (a'_1 \otimes \dots \otimes a'_p)) := (a_1 a'_1, \dots, a_p a'_p).$$

Lemma: Suppose M is an R -module generated by a set B , and $\mu : M \times M \rightarrow M$ is a bilinear map such that $\mu(\mu(b_1, b_2), b_3) = \mu(b_1, \mu(b_2, b_3))$ for all $b_1, b_2, b_3 \in B$. Then $\mu(\mu(m_1, m_2), m_3) = \mu(m_1, \mu(m_2, m_3))$ for all $m_1, m_2, m_3 \in M$.

Proof:

Write $m_1 = \sum_{i=1}^n c_i b_i$, $m_2 = \sum_{j=1}^m c'_j b'_j$, and $m_3 = \sum_{k=1}^\ell c''_k b''_k$. Then:

- $\mu(\mu(m_1, m_2), m_3) = \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^\ell c_i c'_j c''_k \mu(\mu(b_i, b'_j), b''_k),$
- $\mu(m_1, \mu(m_2, m_3)) = \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^\ell c_i c'_j c''_k \mu(b_i, \mu(b'_j, b''_k)).$

And now we just invoke that $\mu(\mu(b_i, b'_j), b''_k) = \mu(b_i, \mu(b'_j, b''_k))$. ■

Based on that lemma, it's now clear that μ defines an associative operation on M . Clearly, μ is the standard multiplication map on $\bigotimes_{i=1}^p A_i$.

Lemma: Suppose A_1, A_2 are R -algebras, A_1 is generated as an R -module by a subset B , $f : A_1 \rightarrow A_2$ is an R -module homomorphism, and $f(bb') = f(b)f(b')$ for all $b, b' \in B$. Then f is an R -algebra homomorphism.

Proof:

We need to show for any $a, a' \in A_1$ that $f(aa') = f(a)f(a')$. Fortunately, we can write $a = \sum_{i=1}^n c_i b_i$ and $a' = \sum_{j=1}^m c'_j b'_j$. Then:

$$\begin{aligned} f(aa') &= f((\sum_{i=1}^n c_i b_i)(\sum_{j=1}^m c'_j b'_j)) = \sum_{i=1}^n \sum_{j=1}^m c_i c'_j f(b_i b'_j) \\ &= \sum_{i=1}^n \sum_{j=1}^m c_i c'_j f(b_i) f(b'_j) \\ &= \left(\sum_{i=1}^n c_i f(b_i) \right) \left(\sum_{j=1}^m c'_j f(b'_j) \right) \\ &= f(\sum_{i=1}^n c_i b_i) f(\sum_{j=1}^m c'_j b'_j) = f(a) f(a'). \quad \blacksquare \end{aligned}$$

Theorem 3.2: Let $A_1, \dots, A_p$ be R -algebras (where $p > 2$) and suppose $1 \leq q < p$. Then the R -module isomorphism $f : (\bigotimes_{i=1}^q A_i) \otimes (\bigotimes_{i=q+1}^p A_i) \rightarrow \bigotimes_{i=1}^p A_i$ given by theorem 2.1 is also an R -algebra homomorphism.

Why?

Just apply the prior lemma where:

$$B = \{(a_1 \otimes \dots \otimes a_q) \otimes (a_{q+1} \otimes \dots \otimes a_p) : a_1, \dots, a_p \in A\}.$$

Theorem 3.4: Let A be an R -algebra and note that R itself is trivially an R -algebra. Thus, we can consider the standard R -algebra structure on $R \otimes A$. I claim that the R -module isomorphism $\phi : R \otimes A \rightarrow A$ satisfying that $\phi(c \otimes a) = ca$ is also an R -algebra homomorphism.

Proof:

Let $B = \{(c \otimes a) : c \in R \text{ and } a \in A\}$. Then:

$$\phi((c \otimes a) \cdot (c' \otimes a')) = \phi(cc' \otimes aa') = cc'aa' = cac'a' = \phi(c \otimes a)\phi(c' \otimes a').$$

So, we can apply the lemma at the bottom of the prior page. ■

Obviously a similar thing holds if we replace $R \otimes A$ with $A \otimes R$.

8/3/2026

Today I can look forward to constructing another important object which I'll mostly try to treat as a black box satisfying a universal property!!

Let M be an R -module. Then suppose A is an R -algebra and $\phi : M \rightarrow A$ is an R -module homomorphism. If B is a second R -algebra and $h : A \rightarrow B$ is a unital R -algebra homomorphism, we know that $h \circ \phi$ is an R -module homomorphism from M to B .

This brings up the following question:

Does there exist a unital R -algebra A and an R -module homomorphism $\phi : M \rightarrow A$ such that for any unital R -algebra B and any R -module homomorphism $f : M \rightarrow B$ there exists a unique unital R -algebra homomorphism $h : A \rightarrow B$ with $h \circ \phi = f$?

$$\begin{array}{ccc} & & A \\ & \nearrow \phi & \vdots \exists! h \\ M & \xrightarrow{f} & B \end{array}$$

If such an R -algebra A and R -module homomorphism ϕ exists then we say A along with ϕ is the tensor algebra of M . Furthermore, if (A_1, ϕ_1) and (A_2, ϕ_2) both satisfy the above universal property then it is easy to see that there exists inverse R -algebra isomorphisms $f : A_1 \rightarrow A_2$ and $g : A_2 \rightarrow A_1$ such that $f \circ \phi_1 = \phi_2$ and $g \circ \phi_2 = \phi_1$. So, assuming a model of the tensor algebra of M exists, it is unique up to isomorphism.

Theorem 4.1: Such a unital R -algebra A and R -module homomorphism $\phi : M \rightarrow A$ exists. Furthermore:

- (a) $\phi : M \rightarrow A$ is an injection.
- (b) A is generated as an R -algebra by $\phi(M)$.
- (c) There is a grading on $\{A_n\}_{n \in \mathbb{Z}}$ on A such that $A_1 = \phi(M)$.
- (d) The grading $\{A_n\}_{n \in \mathbb{Z}}$ is nonnegative and $A_0 = R1_A$.
- (e) For $p \geq 1$, $T_p(M)$ is isomorphic to A_p via the unique R -module homomorphism sending $m_1 \otimes \cdots \otimes m_p$ to $\phi(m_1) \cdots \phi(m_p)$.

Proof:

I'll first focus on just constructing A and ϕ . So let $A_p := T_p(M)$ for all $p \in \mathbb{Z}_{\geq 1}$ and let $A_0 = R$. Then let us set $A := \bigoplus_{p=0}^{\infty} A_p$. Clearly A has an R -module structure. But we still need to define a multiplication operation on A such that A becomes an algebra.

Fortunately for any $p, q \in \mathbb{Z}_{\geq 1}$ we know by theorem 2.1 that there is a (unique) R -module homomorphism $\phi_{p,q} : T_p(M) \otimes T_q(M) \rightarrow T_{p+q}(M)$ such that:

$$\phi_{p,q}((m_1 \otimes \cdots \otimes m_p) \otimes (m'_1 \otimes \cdots \otimes m'_q)) = m_1 \otimes \cdots \otimes m_p \otimes m'_1 \otimes \cdots \otimes m'_q$$

In turn, we know there is a (unique) bilinear map $\mu_{p,q} : T_p(M) \times T_q(M) \rightarrow T_{p+q}(M)$ such that:

$$\mu_{p,q}((m_1 \otimes \cdots \otimes m_p), (m'_1 \otimes \cdots \otimes m'_q)) = m_1 \otimes \cdots \otimes m_p \otimes m'_1 \otimes \cdots \otimes m'_q$$

Similarly for any $p \in \mathbb{Z}_{\geq 1}$ we know there is a (unique) R -module homomorphism $\phi_{0,p} : R \otimes T_p(M) \rightarrow T_p(M)$ such that:

$$\phi_{0,p}(c \otimes (m_1 \otimes \cdots \otimes m_p)) = c(m_1 \otimes \cdots \otimes m_p).$$

In turn, we get a (unique) bilinear map $\mu_{0,p} : R \times T_p(M) \rightarrow T_p(M)$ such that $\mu(c, m_1 \otimes \cdots \otimes m_p) = c(m_1 \otimes \cdots \otimes m_p)$. And we can similarly construct (unique) maps $\mu_{0,0} : R \times R \rightarrow R$ and $\mu_{p,0} : T_p(M) \times R \rightarrow T_p(M)$ for all $p \geq 1$.

I now define multiplication on A as follows. Note that any element $x \in A$ can be written uniquely as $\sum_{p=0}^{\infty} x_p$ where $x_p \in A_p$ for all p and all but finitely many x_p are zero. So, having written $x = \sum_{p=0}^{\infty} x_p$ and $y = \sum_{p=0}^{\infty} y_p$ then I define:

$$\mu(x, y) := \sum_{p=0}^{\infty} \sum_{q=0}^{\infty} \mu_{p,q}(x_p, y_q).$$

- I claim $\mu : A \times A \rightarrow A$ is bilinear. After all, let $c_1, c_2 \in R$ and let $x = \sum_{p=0}^{\infty} x_p$, $y = \sum_{p=0}^{\infty} y_p$, and $z = \sum_{p=0}^{\infty} z_p$ be elements of A . Then:

$$\begin{aligned} \mu(x, c_1 y + c_2 z) &= \sum_{p,q \geq 0} \mu_{p,q}(x_p, c_1 y_q + c_2 z_q) = \sum_{p,q \geq 0} (c_1 \mu_{p,q}(x_p, y_q) + c_2 \mu_{p,q}(x_p, z_q)) \\ &= c_1 \sum_{p,q \geq 0} \mu_{p,q}(x_p, y_q) + c_2 \sum_{p,q \geq 0} \mu_{p,q}(x_p, z_q) \\ &= c_1 \mu(x, y) + c_2 \mu(x, z) \end{aligned}$$

And analogous reasoning shows that $\mu(c_1 x + c_2 y, z) = c_1 \mu(x, z) + c_2 \mu(y, z)$.

- I also claim $\mu(\mu(x, y), z) = \mu(x, \mu(y, z))$ for all $x, y, z \in A$. After all, let $B = \{1_R\} \cup \left(\bigcup_{p=1}^{\infty} \{m_1 \otimes \cdots \otimes m_p \in A_p : m_1, \dots, m_p \in M\} \right)$. Then B generates A as an R -module and its easy to see that:

$$\mu(\mu(b_1, b_2), b_3) = \mu(b_1, \mu(b_2, b_3)) \text{ for all } b_1, b_2, b_3 \in B.$$

So by the lemma on the middle of [page 1081](#), we know this claim holds.

- Also note that $1_R \in A_0$ satisfies that $\mu(1_R, x) = x = \mu(x, 1_R)$. Therefore, upon letting μ be the multiplication operation that A is equipped with, then A is unital with $1_A = 1_R$.

Thus A equipped with μ becomes a unital A -algebra. If we set $A_n = \{0\}$ for all $n < 0$ then it is clear that A is graded by the nonnegative grading $\{A_n\}_{n \in \mathbb{Z}}$. Also $A_0 = R1_A$. And since $M = T_1(M)$, if we let $\phi : M \rightarrow A = R \oplus \bigoplus_{p=1}^{\infty} T_p(M)$ be the embedding map then we have that ϕ is injective and $A_1 = \phi(M)$. Plus, it is easy to see that A is generated as an R -algebra by A_1 and that each A_p is isomorphic to $T_p(M)$ in the way wanted by property (e).

To finish off the main part of the proof, I claim that A and ϕ satisfy the universal property of the tensor algebra of M .

After all, let B an R -algebra and suppose $f : M \rightarrow B$ is an R -module homomorphism. Then note for all $p \geq 1$ that $f^{(p)} : M \times \cdots \times M \rightarrow B$ given by $f^{(p)}(m_1, \dots, m_p) = f(m_1)f(m_2)\cdots f(m_p)$ is also a p -linear map. So, by the universal property of $T_p(M)$ there is a unique R -module homomorphism $\psi^{(p)} : T_p(M) \rightarrow B$ such that:

$$\psi^{(p)}(m_1 \otimes m_2 \otimes \cdots \otimes m_p) = f(m_1)f(m_2)\cdots f(m_p).$$

We also define the R -module homomorphism $\psi^{(0)} : R \rightarrow B$ by $\psi^{(0)}(c) = c1_B$.

By the universal property of coproducts of R -modules, there is a unique R -module homomorphism h from $A = \bigoplus_{p=0}^{\infty} A_p$ to B such that $h(x_p) = \psi^{(p)}(x_p)$ for all $x_p \in A_p$ (where $p \geq 0$). Also, it's clear that $h(1_A) = \psi^{(0)}(1_R) = 1_B$. And we can further verify h is an R -algebra homomorphism via the second lemma on [page 1081](#). After all if $p, q \geq 1$ then:

$$\begin{aligned} h((m_1 \otimes \cdots \otimes m_p) \cdot (m'_1 \otimes \cdots \otimes m'_q)) &= h(m_1 \otimes \cdots \otimes m_p \otimes m'_1 \otimes \cdots \otimes m'_q) \\ &= f(m_1)\cdots f(m_p)f(m'_1)\cdots f(m'_q) \\ &= h(m_1 \otimes \cdots \otimes m_p)h(m'_1 \otimes \cdots \otimes m'_q) \end{aligned}$$

Finally, I claim h is the unique unital R -algebra homomorphism such that:

$$\begin{array}{ccc} & & A \\ & \nearrow \phi & \vdots h \\ M & \xrightarrow{f} & B \end{array}$$

After all, if $h' : A \rightarrow B$ was another unital R -algebra homomorphism that worked then we'd have to have that $h = h'$ on A_1 . Yet since A_1 generates A as a unital R -algebra, we can conclude $h = h'$ everywhere on A .

This finishes showing that there exists a model (A, ϕ) of the tensor algebra of M .

One more note I want to give is that if (A', ϕ') is another model of the tensor algebra of M , then properties (a) through (e) will still hold for A' and ϕ' .

To see why, note that there are unique inverse R -algebra isomorphisms $f : A \rightarrow A'$ and $g : A' \rightarrow A$ satisfying that $f \circ \phi = \phi'$ and $g \circ \phi' = \phi$. And this already guarantees (a) and (b) will hold.

Futhermore, let $A'_n = f(A_n)$ for all $n \in \mathbb{Z}$. Then we can easily see the grading $\{A'_n\}_{n \in \mathbb{Z}}$ on A' will satisfy (c), (d), and (e). ■

For hopefully obvious reasons, we denote the tensor algebra of M as $T(M)$. And if $T(M)$ is equipped with the grading $\{A_n\}_{n \in \mathbb{Z}}$ described in the last theorem then we identify the p th tensor power of M , $T_p(M)$ with A_p for all $p \geq 0$ (where $T_0(M)$ is understood to just be R). In particular, this means we'll identify $M = T_1(M)$ as a submodule of $T(M)$ and $R = T_0(M)$ as a subring of $T(M)$ sharing the same identity element.

Also we'll typically use $\otimes$ as the multiplication sign in $T(M)$.

One notable consequence of having a natural way of viewing M as a submodule of $T(M)$ is as follows:

Suppose M and N are R -modules and $f : M \rightarrow N$ is an R -algebra homomorphism. By viewing N as an R -submodule of $T(N)$, we can then say by the universal property of $T(M)$ that there is a unique unital R -algebra homomorphism $T(f) : T(M) \rightarrow T(N)$ such that $T(f)|_M = f$.

Furthermore, note that $x \in T(M)$ is homogeneous of degree 1 iff $x \in M$. Yet then $(T(f))(x) = f(x) \in N$. So, $(T(f))(x) \in T(N)$ is homogeneous of degree 1 as well and so we can conclude $(T(f))(T_1(M)) \subseteq T_1(N)$.

By lemma 3.1 on [page 1071](#) it then follows that $T(f)$ maps a generating set for $T_p(M)$ (as an R -submodule) into $T_p(N)$ for all $p \geq 2$. Plus, it's obvious that:

$$(T(f))(T_0(M)) = R(T(f))(1_{T(M)}) = R1_{T(N)} = T_0(N).$$

So, $T(f) : T(M) \rightarrow T(N)$ is a graded R -algebra homomorphism.

Another way of describing the prior consequence is as follows. Let $\mathbf{GrAlg}_R$ denote the category whose objects are graded unital R -algebras and whose morphisms are (unital) graded algebra homomorphisms.

We can define a functor $T : R\text{-mod} \rightarrow \mathbf{GrAlg}_R$ mapping any R -module M to its tensor algebra and mapping any R -module homomorphism $f : M \rightarrow N$ to the graded R -algebra homomorphism $T(f) : T(M) \rightarrow T(N)$ described earlier.

Note that $T(\text{Id}_M) = \text{Id}_{T(M)}$ for all R -modules M . After all, $\text{Id}_{T(M)} : T(M) \rightarrow T(M)$ is a graded R -algebra homomorphism whose restriction to M is just the inclusion map $M \hookrightarrow T(M)$. Yet by the universal property of tensor algebras there is only one unital R -algebra homomorphism $T(\text{Id}_M)$ with that property.

In a similar vein, if $f : M \rightarrow N$ and $g : N \rightarrow K$ are R -module homomorphisms, then $T(g \circ f) = T(g) \circ T(f)$. After all, the restriction of either map to M is just the map $g \circ f$. So, both maps must be the same unique map satisfying that. And this proves that $T : R\text{-mod} \rightarrow \mathbf{GrAlg}_R$ is a functor.

One other note is that every element of $T_1(M) = M$ is homogeneous in $T(M)$. Therefore, if $K \subseteq M$ is a submodule then we know from lemma 3.2 on [page 1071](#) that the two-sided ideal $I(K) \subseteq T(M)$ generated by K is homogeneous.

Theorem 4.3: Let $f : M \rightarrow N$ be a surjective R -module homomorphism and let $K = \ker(f)$. Then $T(f) : T(M) \rightarrow T(N)$ is surjective and $\ker(T(f)) = I(K)$ where $I(K)$ is the two-sided ideal in $T(M)$ generated by K .

Proof:

Firstly, f being surjective means that $T_1(N) \subseteq (T(f))(T(M))$. And since $T_1(N)$ is a generating set for $T(N)$ as an R -algebra and we can easily verify that the image of a unital R -algebra homomorphism is a subalgebra, we in turn must have that $(T(f))(T(M)) = T(N)$. So, $T(f)$ is surjective.

Next note that it's obvious that $I(K) \subseteq \ker(T(f))$ since $K \subseteq \ker(f)$ and we can easily verify that the kernel of an R -algebra homomorphism is an ideal and R -module.

As for the other inclusion, consider the quotient map $\pi : T(M) \rightarrow T(M)/I(K)$. We know from [page 1072](#) that $T(M)/I(K)$ has a natural grading $\{A_n\}_{n \in \mathbb{N}}$ satisfying that M/K is isomorphic to A_1 as an R -module via the map $\phi(m+K) := m+I(K)$. Also note by the first isomorphism theorem that $f : M \rightarrow N$ induces an R -module isomorphism $\tilde{f} : M/K \rightarrow N$.

So, $g := \phi \circ \tilde{f}^{-1}$ is an R -module homomorphism from N to $A_1 \subseteq T(M)/I(K)$ satisfying that $g(n) = m + I(K)$ where $m \in M$ is any element satisfying that $f(m) = n$. By the universal property of $T(N)$, we can then extend g to a unique R -algebra homomorphism $T(g) : T(N) \rightarrow T(M)/I(K)$.

But now note that $T(g) \circ T(f) = \pi$. After all, we can easily check for pure tensors that $(T(f))(m_1 \otimes \cdots \otimes m_p) = f(m_1) \otimes \cdots \otimes f(m_p)$. And then if $n_i = f(m_i)$, we have that:

$$\begin{aligned} (T(g))(n_1 \otimes \cdots \otimes n_p) &= g(n_1) \cdots g(n_p) \\ &= (m_1 + I(K)) \cdots (m_p + I(K)) = (m_1 \otimes \cdots \otimes m_p) + I(K). \end{aligned}$$

So, $T(g) \circ T(f)$ agrees with π on a generating set for $T(M)$ as an R -module and it follows that $T(g) \circ T(f) = \pi$ everywhere.

Consequently, $\ker(T(f)) \subseteq \ker(\pi) = I(K)$ and thus we've shown the other inclusion we wanted. ■

As a side note, $T(f)$ restricted to $T_p(M)$ is the map $\otimes^p f := \underbrace{f \otimes \cdots \otimes f}_{p \text{ times}}$.

This is because both maps satisfy that:

$$m_1 \otimes \cdots \otimes m_p \mapsto f(m_1) \otimes \cdots \otimes f(m_p).$$

But also note that we can now conclude that:

$$\ker(\otimes^p f) = I(K) \cap T_p(M) \subseteq T(M)$$

where $K = \ker(f)$ and $I(K)$ is the two-sided ideal in $T(M)$ generated by K .

8/4/2026

Welp I realized my tangent for the last 5 days might have not been necessary after all. I'd still like to think it was informative though since I don't have much experience right now working with R -algebras. I'll start off today though going over two more results about tensor algebras from Northcott's book.

To start off, given a set X , here's how to construct the free R -algebra of X .

Let M be the free R -module of X (equipped with the embedding $\iota_m : X \rightarrow M$). Next, consider the tensor algebra $T(M)$ of M equipped with the R -module homomorphism $\phi : M \rightarrow T(M)$. And finally let $\iota : X \rightarrow T(M)$ be defined by $\iota := \phi \circ \iota_m$. Note that ι will be injective since both ϕ and ι_m are. Also, $\iota(X)$ will generate $T(M)$ as an R -algebra.

Theorem 4.4: If A is a unital R -algebra and $f : X \rightarrow A$ is any function, then there is a unique unital R -algebra homomorphism $\tilde{f} : T(M) \rightarrow A$ such that the following diagram holds:

$$\begin{array}{ccc} & & T(M) \\ & \nearrow \iota & \downarrow \tilde{f} \\ X & \xrightarrow{f} & A \end{array}$$

Proof:

Firstly such an $\tilde{f}$ exists. After all, A is also an R -module, we can conclude by the universal property of M and ι_m that there is a unique R -module homomorphism $g : M \rightarrow A$ such that $g \circ \iota_m = f$. Next, we can use the universal property of $T(M)$ and ϕ to say that there is a unique R -algebra homomorphism $\tilde{f} : T(M) \rightarrow A$ such that $\tilde{f} \circ \phi = g$. And thus $\tilde{f} \circ \iota = (\tilde{f} \circ \phi) \circ \iota_m = g \circ \iota_m = f$.

$$\begin{array}{ccccc} & & & T(M) & \\ & & \nearrow \iota & \downarrow \exists! \tilde{f} & \\ & M & \nearrow \phi & & \\ X & \xrightarrow{\iota_m} & & & A \\ & \searrow \exists! g & \xrightarrow{f} & & \end{array}$$

Next suppose $h : T(M) \rightarrow A$ is another unital R -algebra homomorphism satisfying that $h \circ \iota = f$. Then as $\iota(X)$ generates all of $T(M)$ as an R -algebra, we can say that h is uniquely determined by its restriction to $\iota(X)$. Yet the commuting diagram for $\tilde{f}$ says that $h(\iota(x)) = f(x) = \tilde{f}(\iota(x))$ for all $x \in X$. So, h must equal $\tilde{f}$ everywhere on $T(M)$. ■

As a side note we can also show through standard arguments that if (A, ι) and (A', ι') both satisfy the universal property of a free R -algebra of a set X , then there are inverse R -algebra isomorphisms $f : A \rightarrow A'$ and $g : A' \rightarrow A$ such that $f \circ \iota = \iota'$ and $g \circ \iota' = \iota$.

On a different note, it's easy to see that if M is a free R -module with a linearly independent basis B then $T(M)$ is also a free R -module with a linearly independent basis consisting of all pure tensors whose components are in B .

A related result we can prove is that M is a projective R -module if and only if the tensor algebra $T(M)$ of M is a projective R -module.

Lemma 4.1: Let $u : N \rightarrow M$ and $v : M \rightarrow N$ be R -module homomorphisms such that $v \circ u = \text{Id}_N$. Then u is an injection, v is a surjection, and $M = u(N) \oplus \ker(v)$ as an internal direct sum.

Proof:

That u is an injection and v a surjection is obvious. Also note for any $m \in M$ that $m - u(v(m)) \in \ker(v)$ since:

$$v(m - u(v(m))) = v(m) - v(u(v(m))) = v(m) - v(m) = 0.$$

Similarly $u(v(m)) \in u(N)$. And since $m = u(v(m)) + (m - u(v(m)))$ we can conclude that $u(N) + \ker(v) = M$. Finally, suppose $m \in u(N) \cap \ker(v)$. Then there exists $n \in N$ such that $u(n) = m$. Yet also note that:

$$n = \text{Id}_N(n) = v(u(n)) = v(m) = 0. \text{ So } m = u(0) = 0.$$

And this proves that $u(N) + \ker(v)$ is actually an internal direct sum. ■

Theorem 4.5: Let P be an R -module and let $T(P)$ be the tensor algebra of P . Then P is a projective R -module if and only if $T(P)$ is a projective R -module.

($\Leftarrow$)

Note that $T(P) \cong P \oplus (\bigoplus_{\substack{p \geq 0 \\ p \neq 1}} T_p(P))$ as R -modules.

But now note that $T(P)$ is projective if and only if there is a free R -module $\mathcal{F}$ and some R -module K such that $T(P) \oplus K \cong \mathcal{F}$. In turn:

$$\mathcal{F} \cong (P \oplus (\bigoplus_{\substack{p \geq 0 \\ p \neq 1}} T_p(P))) \oplus K \cong P \oplus ((\bigoplus_{\substack{p \geq 0 \\ p \neq 1}} T_p(P)) \oplus K)$$

So, P is also a direct summand of a free R -module and is thus projective.

As a side note, analogous reasoning lets us further say that if $T(P)$ is a projective R -module then $T_p(P)$ is a projective R -module for all $p \in \mathbb{Z}_{\geq 0}$.

($\Rightarrow$)

Since P is projective we can choose an R -module Q such that $P \oplus Q \cong \mathcal{F}$ where $\mathcal{F}$

is a free R -module. Then we let $\iota : P \hookrightarrow \mathcal{F}$ be the direct sum embedding and we let $\pi : \mathcal{F} \twoheadrightarrow P$ be the direct sum projection. By the universal property of $T(P)$ and $T(\mathcal{F})$, we know that ι and π uniquely extend to R -algebra homomorphisms:

$$T(\iota) : T(P) \rightarrow T(\mathcal{F}) \text{ and } T(\pi) : T(\mathcal{F}) \rightarrow T(P).$$

But now as $\pi \circ \iota = \text{Id}_P$, we can conclude that $T(\pi) \circ T(\iota) = T(\text{Id}_P) = \text{Id}_{T(P)}$. So by the prior lemma, we can conclude that $T(\mathcal{F}) \cong T(P) \oplus \ker(T(\pi))$. Yet also note that since $\mathcal{F}$ is a free R -module, we know $T(\mathcal{F})$ is also a free R -module. So, $T(P)$ is a direct summand of a free R -module and is thus projective. ■

I'm now going to return to what I was trying and failing to construct on 7/27/2026. Also for the sake of continuity I'll go back to the red and orange text tangent color scheme.

My endgoal is to show that if R is a unital commutative ring then the category $\mathbf{CAlg}_R$ (whose objects are unital commutative R -algebras and whose morphisms are unital R -algebra homomorphisms) is cocomplete. There will be three steps to showing this.

1. I need to show $\mathbf{CAlg}_R$ admits *filtered* colimits. In other words if $(I, \lesssim)$ is a directed set (which we turn into a category $\mathbf{I}$ as described on [page 1024](#)), then any functor $F : \mathbf{I} \rightarrow \mathbf{CAlg}_R$ has a colimit.
2. Then I need to use the filtered colimit construction to construct coproducts / tensor products over arbitrary collections of commutative unital R -algebras.
3. Finally, I can adapt the construction on [pages 1023-1024](#) for constructing colimits of R -modules in order to construct general colimits in the category $\mathbf{CAlg}_R$.

Suppose $(I, \lesssim)$ is a directed set, that every $i \in I$ is associated with a commutative unital R -algebra A_i , and that for all $i, j \in I$ with $i \lesssim j$ there exists a unital R -algebra homomorphism $r_j^i : A_i \rightarrow A_j$ such that:

- $i \lesssim j \lesssim k$ in I implies that $r_k^i = r_k^j \circ r_j^i$,
- $r_i^i = \text{Id}_{A_i}$ for all $i \in I$.

In other words, we have a functor $F : I \rightarrow \mathbf{CAlg}_R$. Now here's how to construct $\varinjlim(F)$.

Let $S := \bigcup_{i \in I} (\{i\} \times A_i)$ be the disjoint union of the algebras A_i . Then equip S with the equivalence relation where $(i, x) \sim (j, y)$ if there exists $k \in I$ with $i \lesssim k$ and $j \lesssim k$ such that $r_k^i(x) = r_k^j(y)$. Finally, let $A := S / \sim$ and let $\iota_i : A_i \rightarrow A$ be the map $x \mapsto [(i, x)]$ for all $i \in I$.

If $\mathcal{G} : \mathbf{CAlg}_R \rightarrow \mathbf{Set}$ is the forgetful functor then we know from the construction on [pages 1024-1026](#) that A equipped with all the ι_i is the colimit $\varinjlim(\mathcal{G} \circ F)$.

Thus if we can equip A with a unital R -algebra structure and then show that $\iota_i : A_i \rightarrow A$ is a unital R -algebra homomorphism for all i , then this will be enough to conclude that A equipped with all the ι_i is the colimit $\varinjlim(F)$.

Here's how we define algebra operations on A .

- (Scaling): Given any $c \in R$ and $[(i, x)] \in A$, we let $c[(i, x)] := [(i, cx)]$.
To see that this is well-defined, suppose $[(i, x)] = [(j, y)]$. Then there exists $k \in I$ with $i \lesssim k$, $j \lesssim k$, and $r_k^i(x) = r_k^j(y)$. Then as r_k^i and r_k^j are R -algebra homomorphisms, we have that $r_k^i(c \cdot x) = c \cdot r_k^i(x) = c \cdot r_k^j(y) = r_k^j(c \cdot y)$. As a result, $[(i, cx)] = [(j, cy)]$.
- (Addition): Suppose $[(i, x)], [(j, y)] \in A$. Then given any $k \in I$ with $i \lesssim k$ and $j \lesssim k$ we let $[(i, x)] + [(j, y)] := [(k, r_k^i(x) + r_k^j(y))]$.

To see that this is well-defined, suppose $[(i_1, x_1)] = [(i_2, x_2)]$ and $[(j_1, y_1)] = [(j_2, y_2)]$, and then consider any $k_1, k_2 \in I$ such that $i_1 \lesssim k_1$, $j_1 \lesssim k_1$, $i_2 \lesssim k_2$, and $j_2 \lesssim k_2$. We need to show that:

$$[(k_1, r_{k_1}^{i_1}(x_1) + r_{k_1}^{j_1}(y_1))] = [(k_2, r_{k_2}^{i_2}(x_2) + r_{k_2}^{j_2}(y_2))].$$

Fortunately, note that as $[(i_1, x_1)] = [(i_2, x_2)]$ there exists $\ell \in I$ such that $r_\ell^{i_1}(x_1) = r_\ell^{i_2}(x_2)$. Also without loss of generality we can assume $k_1 \lesssim \ell$ and $k_2 \lesssim \ell$. Similarly, as $[(j_1, y_1)] = [(j_2, y_2)]$ we can find $\ell' \in I$ such that $r_{\ell'}^{j_1}(y_1) = r_{\ell'}^{j_2}(y_2)$. And now I claim without loss of generality that we can just take $\ell = \ell'$ and also assume that $k_1, k_2 \lesssim \ell$.

This is because since I is a directed set, we know there exists

$\ell^{\text{new}} \in I$ with $\ell, \ell', k_1, k_2 \lesssim \ell^{\text{new}}$. And then:

$$r_{\ell^{\text{new}}}^{i_1}(x_1) = (r_{\ell^{\text{new}}}^\ell \circ r_\ell^{i_1})(x_1) = (r_{\ell^{\text{new}}}^\ell \circ r_\ell^{i_2})(x_2) = r_{\ell^{\text{new}}}^{i_2}(x_2)$$

Similarly, we can show $r_{\ell^{\text{new}}}^{j_1}(y_1) = r_{\ell^{\text{new}}}^{j_2}(y_2)$.

So, there exists $\ell \in I$ such that $i_1, i_2, j_1, j_2, k_1, k_2 \lesssim \ell$, $r_\ell^{i_1}(x_1) = r_\ell^{i_2}(x_2)$, and $r_\ell^{j_1}(y_1) = r_\ell^{j_2}(y_2)$. And finally, we can now conclude that:

$$\begin{aligned} r_\ell^{k_1}(r_{k_1}^{i_1}(x_1) + r_{k_1}^{j_1}(y_1)) &= r_\ell^{i_1}(x_1) + r_\ell^{j_1}(y_1) \\ &= r_\ell^{i_2}(x_2) + r_\ell^{j_2}(y_2) = r_\ell^{k_2}(r_{k_2}^{i_2}(x_2) + r_{k_2}^{j_2}(y_2)) \end{aligned}$$

- (Multiplication): Suppose $[(i, x)], [(j, y)] \in A$. Then given any $k \in I$ with $i \lesssim k$ and $j \lesssim k$ we let $[(i, x)][(j, y)] = [(k, r_k^i(x)r_k^j(y))]$.

We can use basically identical reasoning to prove multiplication is well-defined as we used to prove addition is well-defined. Just replace all the addition operations with multiplication operations in the prior reasoning.

Now one major reason why I showed all the algebra operations are well-defined *before* I showed they satisfy the scaling/addition/multiplication axioms for a commutative R -algebra is that we can now always insist on working with whichever elements of an equivalence class are most convenient to work with.

As a general observation, it's easy to see that if $i \lesssim j$ then $[(i, x)] = [(j, r_j^i(x))]$. Also, we can see that $[(i, x)] + [(i, y)] = [(i, x + y)]$ and that $[(i, x)][(i, y)] = [(i, xy)]$. Thus, here's how we can "quickly" prove A is a commutative R -algebra.

Let $[(i, x)], [(j, y)], [(k, z)] \in A$. By picking a common upper bound to i, j , and k in I we can assume without loss of generality that $i = j = k$. And then since A_i is a commutative algebra:

- $([(i, x)][(i, y))][(i, z)] = [(i, xyz)] = [(i, x)][[(i, y)][(i, z)]]$,
- $[(i, x)][[(i, y)] + [(i, z)]] = [(i, xy + xz)] = [(i, x)][(i, y)] + [(i, x)][(i, z)]$,
- $([(i, x)] + [(i, y))][(i, z)] = [(i, xz + xy)] = [(i, x)][(i, z)] + [(i, y)][(i, z)]$,
- $[(i, x)][(i, y)] = [(i, xy)] = [(i, yx)] = [(i, y)][(i, x)]$,
- $[(i, x)] + [(i, y)] = [(i, x + y)] = [(i, y + x)] = [(i, y)] + [(i, x)]$.

Additionally suppose $c_1, c_2 \in R$. Then:

- $c_1(c_2[(i, x)]) = [(i, c_1c_2x)] = c_2c_1[(i, x)]$
- $(c_1 + c_2)[(i, x)] = [(i, (c_1 + c_2)x)] = [(i, c_1x + c_2x)] = c_1[(i, x)] + c_2[(i, x)]$.

Also if $c \in R$ then:

- $(c[(i, x))][(i, y)] = [(i, (cx)y)] = [(i, x(cy))] = [(i, x)](c[(i, y)])$,
- $(c[(i, x))][(i, y)] = [(i, (cx)y)] = [(i, c(xy))] = c([(i, x)][(i, y)])$,
- $c([(i, x)] + [(i, y)]) = [(i, c(x + y))] = [(i, cx + cy)] = c[(i, x)] + c[(i, y)]$.

Finally, we just need to show A has an element 1_A and an element 0_A . So, I define $0_A := [(i, 0_{A_i})]$ and $1_A := [(i, 1_{A_i})]$ where i is any element of I .

To see that this is well-defined, note that $[i, 0_{A_i}] = [j, 0_{A_j}]$ and $[i, 1_{A_i}] = [j, 1_{A_j}]$ for all $i, j \in I$. This is because if we let $k \in I$ be such that $i \lesssim k$ and $j \lesssim k$ then $r_k^i(0_{A_i}) = 0_{A_k}$, $r_k^i(1_{A_i}) = 1_{A_k}$, $r_k^j(0_{A_j}) = 0_{A_k}$, and $r_k^j(1_{A_j}) = 1_{A_k}$.

And now we can easily see that $0_A + [(i, x)] = [(i, x)]$ and $1_A[(i, x)] = [(i, x)]$. Also $[(i, x)] + [(i, -x)] = [(i, 0_{A_i})] = 0_A$. So every element of A has an additive inverse.

It's also now obvious that $\iota_i : A_i \rightarrow A$ is a unital R -algebra homomorphism for all $i \in I$.



With that I've finished showing how to construct a *filtered* colimit.

Remark 1: This is actually all we need in order to study stalks and germs on topological spaces. However, I made the claim earlier that $\mathbf{CAlg}_R$ is a cocomplete category. And if I want to actually justify that assertion, I need to do more work.

Remark 2: The prior construction also works for constructing filtered colimits of R -modules. I mention that because it's much easier to interpret what any element of this construction looks like compared to what any element of the general construction on [pages 1023-1024](#) looks like.

8/5/2026

I have two more constructions to make before I can conclude that $\mathbf{CAlg}_R$ is cocomplete. Firstly, given an arbitrary collection $\{A_\beta\}_{\beta \in B}$ of unital commutative algebras I need to construct their coproduct.